A Final Report

On

ResumeInsight (AI-Powered Resume Analysis and Career Enhancement Platform)

A Project Report Submitted in Partial Fulfillment of the Requirements for the degree of

BACHELOR OF TECHNOLOGY

in

Computer Science & Engineering (AI&ML)

by

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Under the Guidance of

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M.M. ENGINEERING COLLEGE

MAHARISHI MARKANDESHWAR (DEEMED TO BE UNIVERSITY)

(NAAC accredited Grade 'A++' University)
MULLANA, AMBALA, HARYANA

2025--2026

DECLARATION

I hereby declare that this project titled **"ResumeInsight: AI-Powered Resume Analysis and Career Enhancement Platform"** is my original work carried out under the guidance of **Dr. Sonali Goyal**.

This project has not been submitted elsewhere for any degree or diploma. All information presented is genuine and follows academic standards.

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CERTIFICATE

Certified that **Ankit Raj (Roll No.: 11233036)** has successfully carried out the project work presented in this project report entitled **"ResumeInsight: AI-Powered Resume Analysis and Career Enhancement Platform"** for the award of **Bachelor of Technology (Computer Science & Engineering)** from **M.M. Engineering College, Mullana, Ambala** under my guidance.

This project report embodies the results of original work carried out by the student. The work presented in this report has not been submitted elsewhere for the award of any other degree or diploma to this candidate or any other person in this or any other University/Institution.

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ABSTRACT

This project presents the design and development of an advanced web-based application titled **"ResumeInsight: AI-Powered Resume Analysis and Career Enhancement Platform."** The primary objective of this system is to assist job seekers in improving their resumes and enhancing their chances of securing employment in a highly competitive job market.

Traditional resume evaluation methods often lack accuracy, personalization, and intelligent feedback mechanisms. Most existing tools rely on basic keyword matching and fail to provide meaningful insights regarding resume quality, skill gaps, and job compatibility. As a result, many candidates are unable to optimize their resumes according to industry standards and Applicant Tracking System (ATS) requirements.

To overcome these limitations, the proposed system integrates advanced Artificial Intelligence technologies using the **Google Gemini 2.5 Flash API**, which enables intelligent analysis of resumes and job descriptions. The system utilizes Natural Language Processing (NLP) techniques to understand resume content, extract relevant skills, and evaluate their alignment with specific job roles. It provides an ATS compatibility score, identifies missing or weak skills, and generates personalized recommendations for improvement.

The application is developed using **Flask as the backend framework**, ensuring efficient handling of user requests and AI processing. The frontend is designed using **HTML, CSS, JavaScript, and Tailwind CSS**, offering a responsive and user-friendly interface. The system also supports multi-format resume uploads (PDF, DOCX, and TXT) and generates detailed reports in various formats.

In addition to resume analysis, ResumeInsight includes several advanced features such as interview preparation tools, job application tracking, salary insights, and career guidance modules. These features collectively provide a complete career support system for users.

By combining AI-powered analysis with practical career tools, the proposed system significantly enhances user experience, improves resume quality, and supports informed decision-making in career development. The project demonstrates the effective application of artificial intelligence in solving real-world problems related to recruitment and employability.

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my project guide, **Dr. Sonali Goyal**, Associate Professor, Department of Computer Science & Engineering, M.M. Engineering College, Mullana, for her continuous support, valuable guidance, and encouragement throughout the development of this project. Her insightful suggestions and constant motivation played a crucial role in the successful completion of this work.

I am truly thankful for the time and effort she invested in guiding me at every stage of the project, from the initial concept to the final implementation. Her expertise and knowledge helped me overcome various challenges and improve the overall quality of the project.
I would also like to thank all the faculty members of the Computer Science & Engineering Department for their support and for providing the necessary resources to complete this project successfully.

Finally, I would like to express my heartfelt gratitude to my family and friends for their encouragement and support throughout this journey.

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CHAPTER 1: INTRODUCTION

1.1 General Overview

In today's digital era, artificial intelligence (AI) is playing a vital role in transforming various industries, including recruitment and career development. Just like having a smart assistant that helps in performing daily tasks efficiently, AI-based systems can now assist job seekers in improving their resumes and enhancing their career opportunities. Instead of manually editing resumes and guessing what recruiters expect, users can now rely on intelligent systems to guide them.

In modern recruitment processes, most companies use Applicant Tracking Systems (ATS) to filter resumes before they reach human recruiters. These systems scan resumes based on keywords, structure, and relevance to the job role. As a result, many candidates fail to pass this stage, not because they lack skills, but because their resumes are not properly optimized.

The proposed system, **ResumeInsight**, is designed as an AI-powered career assistant that helps users analyze and improve their resumes. It provides intelligent feedback, evaluates resume quality, and compares it with job descriptions to identify strengths and weaknesses. The system uses advanced Artificial Intelligence and Natural Language Processing (NLP) techniques to understand resume content and generate meaningful insights.

ResumeInsight supports multiple functionalities such as resume analysis, ATS scoring, skill gap identification, and personalized recommendations. It also includes additional features like interview preparation, job application tracking, and salary insights. The system is developed using Python and Flask for backend processing and uses modern web technologies such as HTML, CSS, JavaScript, and Tailwind CSS for the frontend interface.

By integrating these features, ResumeInsight acts as a smart career companion that simplifies the job search process, improves resume quality, and helps users make better career decisions.

1.2 Problem Statement

Traditional resume evaluation methods and existing tools often suffer from several limitations. Most systems rely on simple keyword matching and fail to understand the actual meaning and context of the resume content. As a result, they provide generic feedback that is not useful for users.

One of the major problems faced by job seekers is the inability to pass Applicant Tracking Systems (ATS). Many candidates are unaware of how ATS works, which leads to poorly formatted resumes that get rejected automatically. Additionally, candidates often do not know

which skills are required for a specific job role, resulting in a mismatch between their profile and job expectations.

Another challenge is the lack of a unified platform that provides complete career support. Users have to rely on different tools for resume building, job tracking, interview preparation, and salary insights. This makes the process inefficient and time-consuming.

Users also face difficulties such as:

- Inaccurate or generic feedback that does not provide clear improvement suggestions.
- Lack of skill gap analysis and career guidance.
- No proper system to track job applications and progress.
- Complex interfaces that are difficult to use.

To overcome these challenges, ResumeInsight is developed as an intelligent, interactive, and user-friendly platform. It integrates AI technology using the Google Gemini API along with NLP techniques to analyze resumes effectively. The system provides accurate insights, identifies missing skills, and offers personalized recommendations, thereby improving the overall user experience and increasing the chances of job selection.

1.3 Objectives

The main objective of this project is to develop an AI-powered system that can analyze resumes and provide intelligent feedback to users. The system aims to help job seekers understand how well their resumes match job requirements and what improvements are needed.

The specific objectives of the system are as follows:

• To develop an AI-based mechanism that analyzes resume content and provides an ATS compatibility score.

• To design a system that compares resumes with job descriptions and identifies matching and missing skills.

• To implement a feature that provides personalized suggestions for improving resume quality and content.

• To create tools for interview preparation, including question generation and guidance.

• To develop a job application tracking system that helps users manage their job search process efficiently.

• To provide salary insights and career guidance based on industry standards.

• To build a user-friendly interface that supports easy interaction and real-time feedback.

1.4 System Modules Overview

The ResumeInsight system is divided into multiple modules, each responsible for performing specific tasks. These modules work together to ensure smooth operation and efficient processing.

AI-Based Resume Analysis Module

- This module uses Google Gemini API and NLP techniques to analyze resume content.
- It evaluates resume quality, extracts important information, and generates an ATS score.
- It provides detailed feedback and improvement suggestions.

Skill Gap Analysis Module

- This module identifies the difference between the user's skills and job requirements.
- It highlights missing skills and suggests ways to improve them.
- It helps users align their profile with industry demands.

AI Tools Module

- This module provides advanced features such as resume enhancement, cover letter generation, and interview question preparation.
- It uses AI to generate personalized and context-aware content.

Job Application Tracking Module

- This module allows users to track their job applications.
- It manages application status, follow-up dates, and company details.
- It helps users stay organized during their job search.

Backend Processing Module

- The backend is developed using Flask, which handles user requests and data processing.
- It communicates with the AI model and manages system operations efficiently.

Database Management Module

- The system uses SQLite to store user data, resumes, and analysis results.
- It ensures secure and efficient data management.

Graphical User Interface (GUI) Module

- The frontend is developed using HTML, CSS, JavaScript, and Tailwind CSS.
- It provides an interactive and user-friendly interface.
- Users can upload resumes, view results, and interact with the system easily.

CHAPTER 2: Literature Review

2.1 Resume Analysis Technology

Resume analysis technology has evolved significantly in recent years with the advancement of Artificial Intelligence (AI) and Natural Language Processing (NLP). Earlier, resume screening was performed manually by recruiters, which was time-consuming and prone to human error. With the increasing number of job applications, organizations have adopted automated systems such as Applicant Tracking Systems (ATS) to filter resumes efficiently.

Modern resume analysis systems utilize AI-based techniques to understand the structure, content, and relevance of resumes. These systems are capable of extracting key information such as skills, experience, education, and keywords, and comparing them with job requirements. Unlike traditional keyword-based systems, advanced AI models can analyze context and provide meaningful insights.

Key Technologies Used in Resume Analysis

- **Natural Language Processing (NLP):** NLP plays a crucial role in understanding resume content. It helps in extracting meaningful information from unstructured text, identifying skills, and analyzing the relevance of the resume with respect to job descriptions.
- **Machine Learning (ML):** Machine learning algorithms enable the system to learn from historical data and improve accuracy over time. These models help in predicting candidate-job fit and improving recommendation quality.
- **Artificial Intelligence Models (LLMs):** Large Language Models such as Google Gemini are used to generate intelligent insights, provide suggestions, and perform contextual analysis of resumes.
- **Document Processing Tools:** Libraries such as PyPDF2 and python-docx are used to extract text from resume files in different formats like PDF and DOCX.
- **API Integration:** Integration with AI APIs allows the system to process data efficiently and generate real-time responses and recommendations.

These technologies collectively contribute to building intelligent resume analysis systems that go beyond simple keyword matching and provide actionable insights.

2.2 Existing AI-Based Resume Tools and Assistants

With the growth of AI technology, several platforms have been developed to assist users in resume building and analysis. These systems provide basic functionalities but often lack advanced features and deep analysis capabilities.

A. LinkedIn Resume Builder

LinkedIn provides a resume builder tool that helps users create professional resumes based on their profiles. It offers suggestions for improving content and formatting. However, it mainly focuses on resume creation rather than detailed analysis or skill gap identification.

B. Indeed Resume Tools

Indeed offers resume building and job matching services. It allows users to upload resumes and apply for jobs directly. While it provides job recommendations, it lacks advanced AI-based analysis and personalized feedback mechanisms.

C. ATS Resume Checkers

Various online tools provide ATS compatibility checks by scanning resumes for keywords and formatting. These tools help users improve their chances of passing automated screening systems. However, most of them rely on basic algorithms and do not provide deep insights or career guidance.

D. AI-Based Assistants (e.g., ChatGPT, Gemini)

AI assistants such as ChatGPT and Google Gemini provide advanced conversational capabilities and can assist users in generating resume content, cover letters, and career advice. These systems use large language models to understand context and generate human-like responses. However, they are not specifically designed as complete resume analysis platforms with integrated tracking and reporting features.

Although these existing systems provide useful functionalities, they are often limited in terms of integration, personalization, and comprehensive career support.

2.3 Challenges in AI-Based Resume Analysis Systems

Despite the advancements in AI-based resume analysis systems, several challenges still affect their efficiency, accuracy, and usability. Addressing these challenges is essential for building a reliable and effective system.

a. Natural Language Understanding Limitations

AI systems sometimes face difficulty in understanding complex or ambiguous resume content. Variations in writing style, formatting, and terminology can impact the accuracy of analysis. Additionally, handling context across multiple sections of a resume remains a challenge.

b. Accuracy and Relevance of Suggestions

While AI models can generate recommendations, they may sometimes produce generic or less relevant suggestions. Ensuring that the feedback is accurate, personalized, and aligned with industry requirements is a key challenge.

c. Skill Extraction and Matching Issues

Identifying relevant skills from resumes and matching them with job descriptions requires precise analysis. Inaccurate extraction or incorrect matching can lead to misleading results and poor recommendations.

d. Data Privacy and Security

Resume analysis systems handle sensitive user data such as personal details, education, and work experience. Ensuring data privacy, secure storage, and safe API communication is essential to maintain user trust.

e. Integration and Scalability

Integrating multiple technologies such as AI APIs, databases, and frontend interfaces can be complex. Ensuring smooth communication between different components and maintaining system performance at scale is a significant challenge.

f. User Experience and Interface Design

Many systems fail to provide a user-friendly interface, making it difficult for users to understand analysis results. A clear, interactive, and intuitive interface is necessary to enhance usability and engagement.

By analyzing the limitations of existing systems and challenges in current technologies, ResumeInsight is designed to provide a more comprehensive, intelligent, and user-friendly solution. It integrates advanced AI models, real-time analysis, and multiple career support features to overcome these issues and deliver an enhanced user experience.

CHAPTER 3: Design and Implementation

3.1 Software Design

The ResumeInsight system is designed using a modular and scalable architecture to ensure flexibility, efficiency, and ease of maintenance. The system is structured in such a way that different components work independently while maintaining seamless communication with each other. This modular design approach helps in improving system performance and allows future enhancements without affecting existing functionalities.

The system consists of several core components that together form the complete application:

• User Interface (UI): The frontend of the system is developed using HTML, CSS, JavaScript, and Tailwind CSS. It provides a responsive and user-friendly interface where users can upload resumes, enter job descriptions, and view analysis results. The interface is designed to be simple, intuitive, and accessible for all types of users.

• Backend Processing: The backend is implemented using the Flask framework in Python. It handles user requests, processes resume data, communicates with the AI model, and manages system logic. Flask ensures efficient routing, data handling, and integration with external APIs.

• AI Processing Module: This is the core component of the system, which uses the Google Gemini 2.5 Flash API for intelligent resume analysis. It processes the resume content, performs job matching, and generates meaningful insights such as ATS score, skill gaps, and recommendations.

• Document Processing Module: This module is responsible for extracting text from uploaded resumes. It uses libraries such as PyPDF2 and python-docx to handle different file formats like PDF, DOCX, and TXT. The extracted text is then passed to the AI module for analysis.

• Database Management: The system uses SQLite along with SQLAlchemy ORM to store user data, resume information, analysis results, and job tracking details. This ensures efficient data storage, retrieval, and management.

This modular architecture enables ResumeInsight to perform multiple operations efficiently while maintaining a smooth and responsive user experience.

3.2 Natural Language Processing (NLP)

Natural Language Processing (NLP) plays a crucial role in the ResumeInsight system as it enables the system to understand, interpret, and analyze human language present in resumes and job descriptions. Since resumes are written in unstructured text format, NLP techniques are required to extract meaningful information and convert it into structured data.
The system uses NLP techniques to identify key elements such as skills, experience, education, and keywords from resumes. It also analyzes job descriptions to understand the requirements

and expectations of employers. By comparing both, the system determines how well a resume matches a particular job role.

Technologies Used:

- Tokenization: This process breaks down the resume text into smaller units such as words and phrases, making it easier to analyze.
- Keyword Extraction: Important keywords related to skills, technologies, and experience are identified and categorized.
- Context Analysis: The system understands the meaning and relevance of words within the context of the resume.
- Text Normalization: This includes converting text to a standard format, removing unnecessary characters, and correcting spelling errors.
- Named Entity Recognition (NER): Helps in identifying specific entities such as names, organizations, and job roles.

By applying these NLP techniques, the system is able to perform accurate analysis and generate meaningful insights. This significantly improves the quality of feedback provided to users and enhances overall system performance.

3.3 AI Model Integration (Google Gemini API)

To provide intelligent and context-aware analysis, ResumeInsight integrates the Google Gemini 2.5 Flash AI model. This model is capable of understanding complex text data and generating human-like responses based on the input provided.
The AI model is used to analyze resumes, compare them with job descriptions, and generate detailed insights. It evaluates various parameters such as relevance, skill match, experience level, and formatting quality.

Implementation Steps:

1. The user uploads a resume and provides a job description.
2. The system extracts text from the resume using document processing tools.
3. The extracted data is processed using NLP techniques.
4. The processed input is sent to the Google Gemini API.
5. The AI model analyzes the data and generates results such as ATS score, skill gap analysis, and recommendations.
6. The response is sent back to the frontend and displayed to the user.

Advantages of AI Integration:

- Provides accurate and context-aware analysis.
- Generates personalized recommendations.
- Improves decision-making for users.
- Enhances system intelligence and automation.

By integrating advanced AI technology, ResumeInsight delivers high-quality analysis and meaningful insights that go beyond traditional systems.

3.4 Flask Backend and API Handling

The backend of ResumeInsight is developed using Flask, which is a lightweight and efficient web framework in Python. Flask is responsible for handling user requests, managing application logic, and integrating different components of the system.
Key Responsibilities of Flask Backend:

- Handles user authentication and session management.
- Processes resume uploads and extracts data.
- Communicates with the AI model for analysis.
- Manages database operations using SQLAlchemy.
- Sends processed data and results to the frontend.

API Handling Workflow:

1. The user interacts with the frontend and submits data (resume or job description).
2. The request is sent to the Flask server through API endpoints.
3. Flask processes the request and forwards the data to the appropriate module (NLP or AI processing).
4. The AI model generates the response.
5. Flask receives the response and sends it back to the frontend.

This structured workflow ensures smooth communication between different components and enables real-time processing of user requests.

3.5 HTML, CSS, and JavaScript for GUI

The Graphical User Interface (GUI) of ResumeInsight is designed using modern web technologies such as HTML, CSS, JavaScript, and Tailwind CSS. The main objective of the GUI is to provide a simple, interactive, and visually appealing interface for users.

Features of the GUI:

- Supports resume upload in multiple formats.
- Allows users to input job descriptions.
- Displays analysis results in a clear and structured format.
- Provides real-time feedback and suggestions.
- Includes interactive elements such as dashboards and charts.

Implementation Technologies:

- HTML: Used for structuring the web pages and content layout.
- CSS and Tailwind CSS: Used for styling, responsiveness, and modern design.
- JavaScript: Handles user interactions, form validation, and communication with the backend APIs.
- Chart.js: Used for displaying analytics and visual data representation.

The GUI is designed to be responsive and user-friendly, ensuring that users can easily interact with the system and understand the analysis results without any difficulty.

CHAPTER 4: Results and Analysis

4.1 Screenshots and Output Analysis

Working:
The ResumeInsight system was successfully developed and tested with various user inputs. The application provides a complete workflow starting from resume upload to AI-based analysis and report generation. The following section describes the working of different modules along with their outputs.

Fig. 4.1 Dashboard Interface
The dashboard serves as the main entry point of the system. It provides an overview of user activity, including total analyses performed, average resume score, and recent activity. As shown in the dashboard, the system displays metrics such as total analyses (38) and average score (76.74%), which helps users track their performance over time.

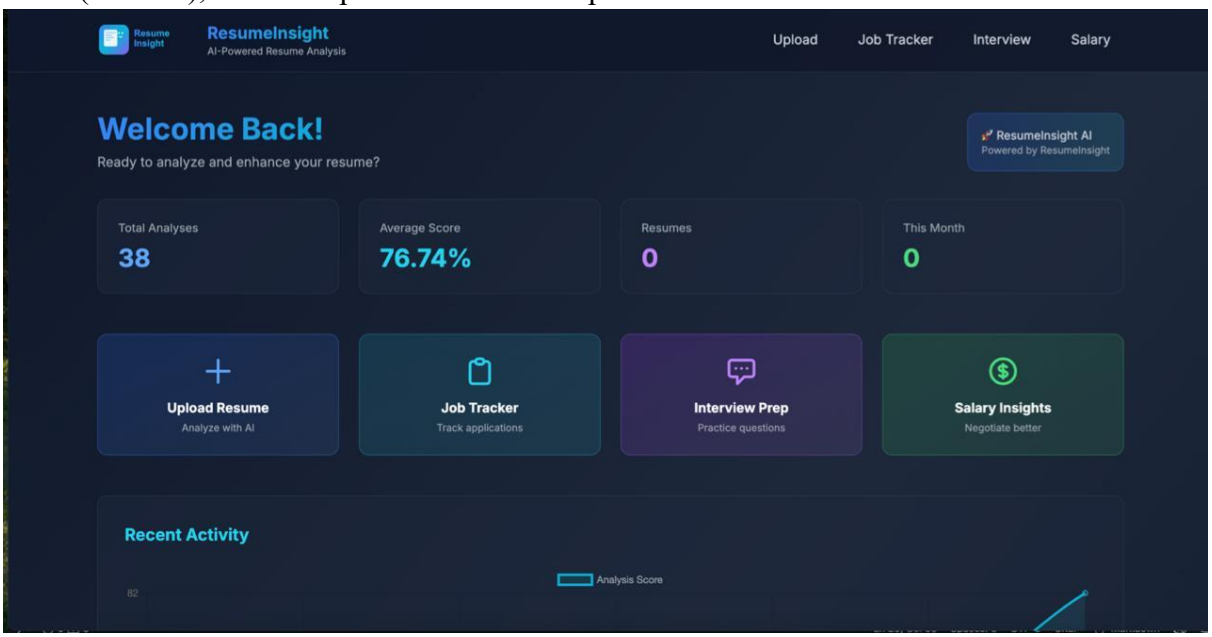


Fig – 4.1

The dashboard also provides quick access to major modules such as Resume Upload, Job Tracker, Interview Preparation, and Salary Insights. This centralized interface improves usability and allows users to navigate the system efficiently.

Fig. 4.2 Interview Preparation Module
The Interview Preparation Hub is designed to help users prepare for job interviews effectively. It follows a structured workflow:
Step 1: Upload Resume
Step 2: Add Job Details

Step 3: Generate Questions
Step 4: Practice and Improve

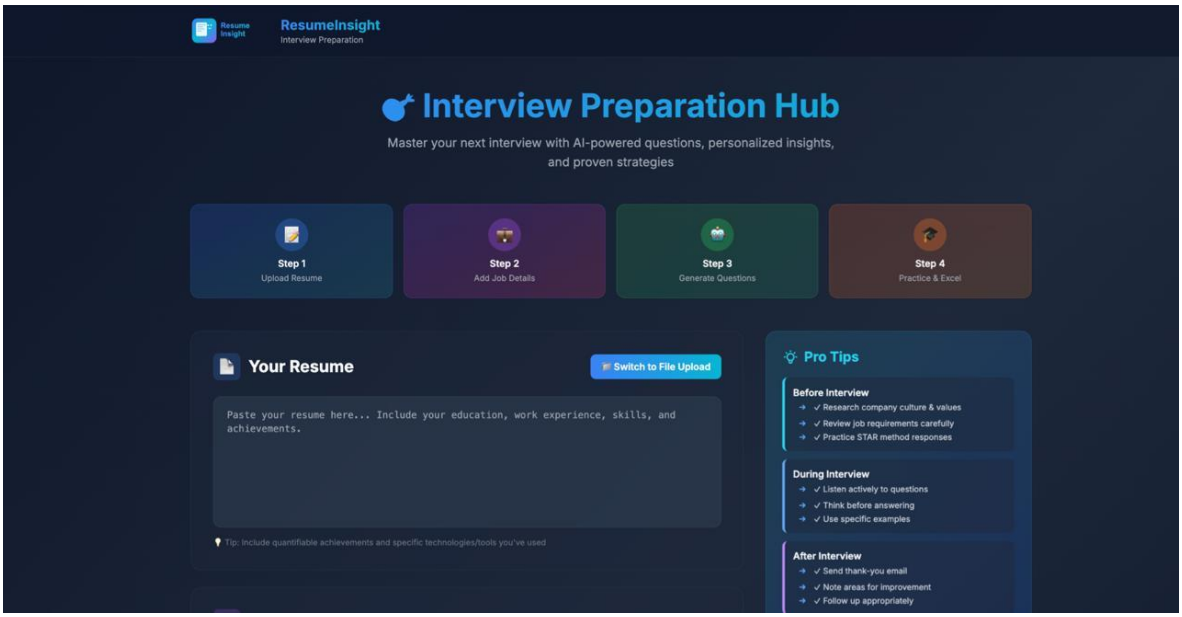


Fig – 4.2

The system generates AI-based interview questions based on the user's resume and job role. Additionally, it provides professional tips categorized into before, during, and after interview stages. This module enhances user confidence and improves interview readiness.

Fig. 4.3 Salary Insights and Negotiation Module

The Salary Insights module provides users with detailed salary analysis based on their resume, job role, experience level, and industry. Users can input details such as target job title, location, and experience level to receive accurate salary predictions.

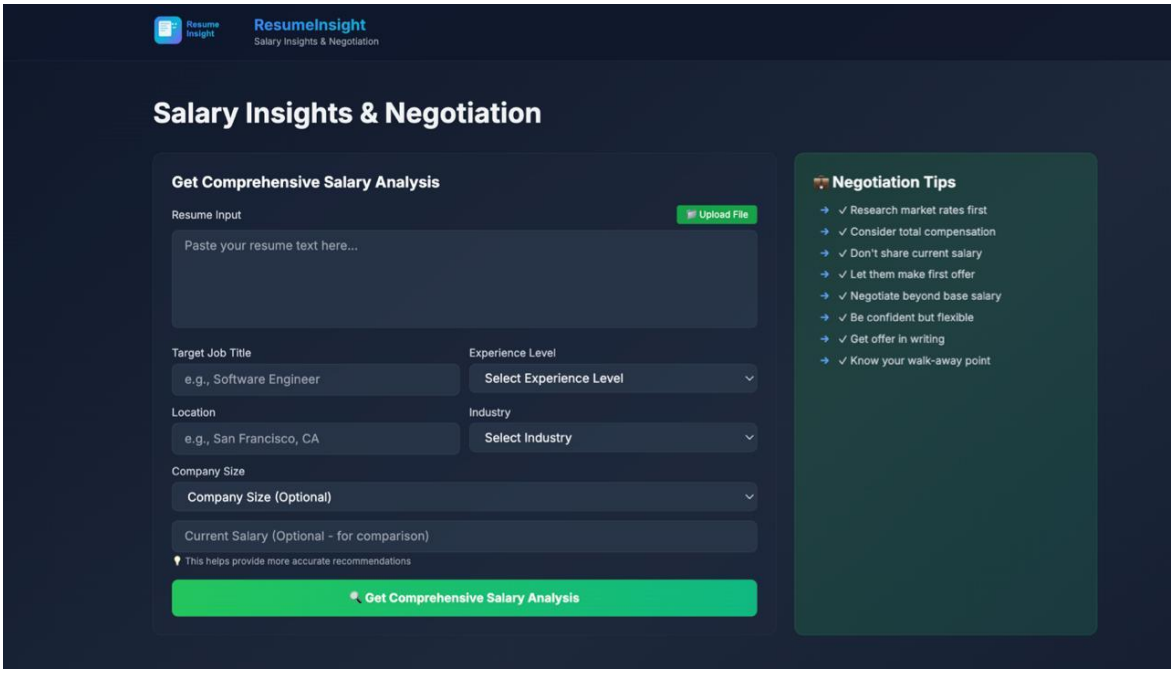


Fig – 4.3

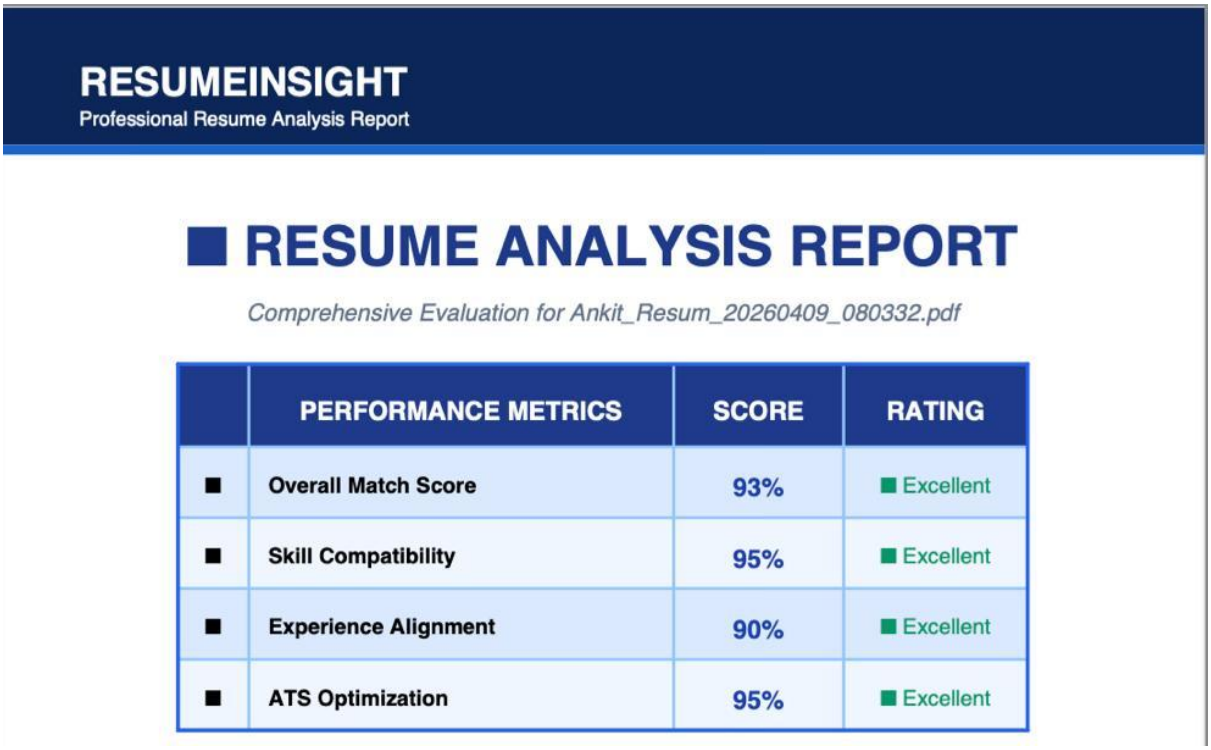
The system also provides negotiation tips such as researching market rates, understanding total compensation, and negotiating beyond base salary. This module helps users make informed financial decisions during job offers.

Fig. 4.4 Resume Analysis Report

The system generates a comprehensive resume analysis report in PDF format. As shown in the report:

• Overall Match Score: 93% (Excellent)
• Skill Compatibility: 95% (Excellent)
• Experience Alignment: 90% (Excellent)
• ATS Optimization: 95% (Excellent)

These metrics indicate how well the resume matches job requirements. The report also includes an executive summary that highlights the candidate's strengths and overall profile evaluation.



RESUMEINSIGHT
Professional Resume Analysis Report

RESUME ANALYSIS REPORT

Comprehensive Evaluation for Ankit_Resum_20260409_080332.pdf

	PERFORMANCE METRICS	SCORE	RATING
■	Overall Match Score	93%	■ Excellent
■	Skill Compatibility	95%	■ Excellent
■	Experience Alignment	90%	■ Excellent
■	ATS Optimization	95%	■ Excellent

Fig – 4.4

Fig. 4.5 Key Strengths and Areas for Improvement

The system identifies key strengths such as strong foundation in AI/ML, practical project experience, and diverse technical skills. At the same time, it highlights areas for improvement such as lack of cloud experience, limited exposure to advanced deep learning architectures, and absence of PyTorch skills.

■ KEY STRENGTHS

1. • Strong Foundation in AI/ML: Demonstrated through academic focus (AI & ML degree), relevant projects, and proficiency in core libraries (TensorFlow, Scikit-learn, XGBoost, LightGBM).
2. • Practical Project Experience: Developed and deployed full-stack AI/ML applications (AI Health Assistant, ResumeInsight, Loan Eligibility Prediction), showcasing end-to-end development capabilities.
3. • Diverse Technical Skillset: Proficient in multiple programming languages (Python, Java), ML/DL frameworks, web frameworks (Flask, Django), and tools (Docker, Git, SQL databases).
4. • High Academic Achievement: Consistent strong CGPA (8.53 B.Tech, 8.75 Diploma) indicates strong learning ability and dedication.
5. • Web Development Integration: Ability to integrate AI/ML models into functional web applications using Flask, Jinja2, HTML/CSS/JS, and database skills.

■■ AREAS FOR IMPROVEMENT

1. • Limited Explicit Cloud Experience: While Google Colab is mentioned, explicit experience with major cloud platforms (AWS, GCP, Azure) for MLOps or scalable deployment is not highlighted.
2. • Future-Dated Internship: The listed internship is in the future (May 2025 - June 2025), meaning there is no current or past professional AI/ML internship experience.
3. • No Specific Deep Learning Architecture Experience: While TensorFlow is listed, specific experience with architectures like CNNs, RNNs, or Transformers for complex tasks is not detailed in projects.
4. • Lack of PyTorch Proficiency: TensorFlow is mentioned, but PyTorch is another dominant deep learning framework not explicitly listed.
5. • Limited MLOps/Deployment Beyond Flask/Docker: While Flask and Docker are good for local deployment, mention of MLOps tools, CI/CD for ML, or orchestrators like Kubernetes is absent.

Fig – 4.5

This balanced analysis helps users understand both their strengths and weaknesses.

Fig. 4.6 AI-Powered Recommendations

The system provides actionable recommendations to improve the resume. These include:

• Gaining cloud platform experience (AWS, GCP, Azure)
• Learning PyTorch framework
• Improving project descriptions with measurable outcomes
• Exploring advanced deep learning techniques

These suggestions are generated using AI and are tailored to the user's profile.

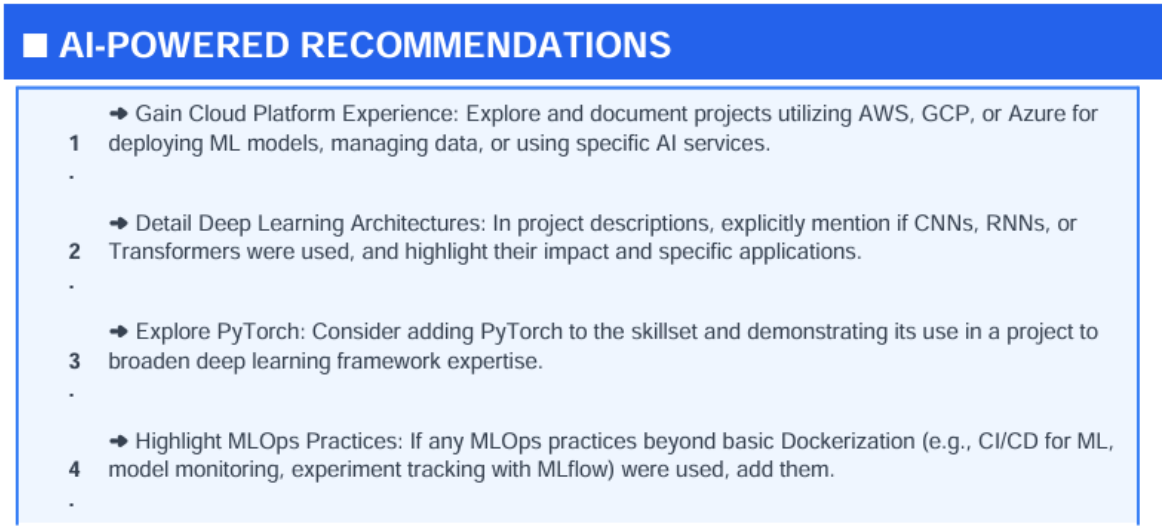


Fig – 4.6

Fig. 4.7 Skill Analysis

The system performs detailed skill analysis by categorizing skills into matched skills and missing skills.

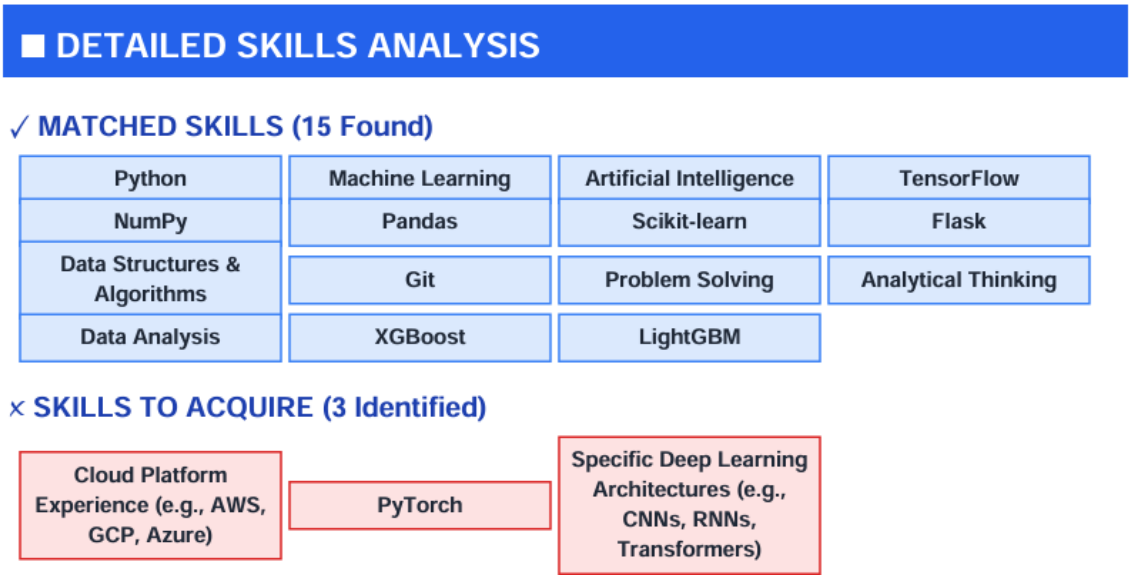


Fig – 4.7

Matched skills include Python, Machine Learning, TensorFlow, Flask, and Data Analysis. Missing skills include cloud platforms, PyTorch, and advanced deep learning architectures. This helps users focus on skill development and improve their employability.

Fig. 4.8 Career Path and Learning Plan

The system provides career insights such as experience level, job match percentage, and career stage. It also suggests a learning path with priority levels and estimated time required for skill development.

■ CAREER PATH ANALYSIS

Assessment Category	Result	Description
Experience Level	Senior Level	Extensive experience demonstrated with advanced skills
Job Match	95% Compatible	Based on skills alignment
Career Stage	Mid-Career	Current professional stage

■ Career Progression

1. Consider senior-level positions or team lead roles
2. Explore specialization in your domain
3. Look into mentorship or consulting opportunities

Fig – 4.8

For example, cloud platform experience is marked as a critical skill, while PyTorch and deep learning architectures are marked as high priority.

Fig. 4.9 Interview Questions and Preparation

The system generates personalized interview questions based on the user's profile. It also provides a structured approach using the STAR method (Situation, Task, Action, Result) to help users answer behavioral questions effectively.

This feature enhances interview performance and helps users prepare in a structured manner.

■ INTERVIEW PREPARATION GUIDE

■ Potential Interview Questions

Q1: Can you describe your experience working with Python?

Q2: Tell me about a challenging project where you used Python.

Q3: Can you describe your experience working with Machine Learning?

Q4: Tell me about a challenging project where you used Machine Learning.

Q5: Can you describe your experience working with Artificial Intelligence?

Q6: Tell me about a challenging project where you used Artificial Intelligence.

Q7: Tell me about a time when you faced a difficult challenge at work and how you overcame it.

Q8: Describe a situation where you had to work with a difficult team member.

Q9: Give an example of a goal you set and how you achieved it.

Q10: How do you prioritize tasks when managing multiple projects?

Fig – 4.9

4.2 Performance and Analysis

The performance of the ResumeInsight system was evaluated based on accuracy, response time, usability, and overall effectiveness.

The system demonstrates high accuracy in analyzing resumes and identifying relevant skills. The AI model successfully generates meaningful insights and personalized recommendations. The response time of the system is fast, ensuring real-time analysis and feedback.

The user interface is intuitive and easy to use, allowing users to interact with the system without any technical difficulty. The integration of multiple modules such as resume analysis, job tracking, and interview preparation makes the system highly efficient and practical.

Overall, the ResumeInsight system performs effectively and provides a comprehensive solution for resume optimization and career enhancement.

Chapter 5: Conclusion and Scope for Improvement

5.1 Summary of Findings

This project aimed to develop an intelligent web-based application titled **"ResumeInsight: AI-Powered Resume Analysis and Career Enhancement Platform."** The primary goal of the system was to assist job seekers in improving their resumes and enhancing their chances of securing employment by leveraging advanced Artificial Intelligence technologies.

The system was successfully designed and implemented using modern technologies such as Flask for backend development, Natural Language Processing (NLP) techniques for text analysis, and the Google Gemini 2.5 Flash API for intelligent resume evaluation. The platform provides a comprehensive solution that not only analyzes resumes but also offers personalized career guidance and improvement suggestions.

One of the key findings of this project is that the integration of AI models significantly improves the accuracy and effectiveness of resume analysis. The system is capable of understanding resume content, extracting relevant information, and comparing it with job descriptions to generate meaningful insights such as ATS compatibility score and skill gap analysis.

Another important observation is that the use of a web-based graphical user interface developed using HTML, CSS, JavaScript, and Tailwind CSS enhances user experience. The interface allows users to easily upload resumes, view analysis results, and interact with the system in a simple and intuitive manner.

The system also successfully integrates multiple functionalities such as job application tracking, interview preparation, and salary insights, making it a complete career support platform rather than just a resume analysis tool. Additionally, the use of document processing libraries ensures smooth handling of different file formats like PDF and DOCX.

Overall, the project demonstrates that combining artificial intelligence with modern web technologies can significantly improve the job application process. It provides users with accurate insights, actionable recommendations, and a structured approach to career development.

5.2 Scope for Improvement

Although the ResumeInsight system has been successfully implemented and provides valuable features, there are still several areas where improvements can be made to enhance its performance, accuracy, and usability.

• Enhanced NLP Processing: While the current system uses NLP techniques to analyze resumes, further improvements can be made by fine-tuning models with domain-specific

datasets. This will enable the system to provide more accurate and context-aware analysis for specialized fields such as software development, healthcare, finance, and management. Advanced NLP techniques such as semantic analysis and deep learning-based models can further improve understanding of resume content.

• Improved AI Accuracy: Although the integration of the Google Gemini API provides high-quality results, there may still be cases where the generated suggestions are generic. Enhancing the system with additional training data and custom AI models can improve the relevance and personalization of recommendations.

• Multilingual Support: Currently, the system primarily supports English language resumes. Expanding support to multiple languages such as Hindi, French, and Spanish will make the platform accessible to a wider audience. This can be achieved by integrating translation APIs and multilingual NLP models.

• Advanced Skill Analysis: The system can be enhanced by incorporating more detailed skill evaluation, including proficiency levels, certification validation, and industry-based skill ranking. This will help users better understand their strengths and weaknesses.

• Data Privacy and Security: Since the system handles sensitive user data, implementing advanced security measures such as encryption, secure authentication, and data anonymization will improve user trust and system reliability.

• Performance Optimization: As the number of users increases, optimizing system performance and ensuring scalability will be important. This can be achieved by using cloud-based infrastructure and efficient API handling techniques.

5.3 Future Work

The ResumeInsight system has significant potential for future enhancements and expansion. Several advanced features can be implemented to further improve its functionality and usability.

• AI-driven Personalization: Future versions of the system can include machine learning models that learn from user behavior and provide personalized recommendations based on individual preferences and career goals.

• Integration with Job Portals: The system can be integrated with platforms such as LinkedIn, Naukri, and Indeed to allow users to directly apply for jobs and receive real-time job recommendations.

• Mobile Application Development: Developing a mobile application for Android and iOS will increase accessibility and allow users to use the platform anytime and anywhere.

• Video Resume Analysis: Future enhancements can include the ability to analyze video resumes using computer vision and AI techniques, providing feedback on communication skills and presentation.

• Advanced Analytics Dashboard: The system can include predictive analytics to forecast career growth, salary trends, and job opportunities based on user data.

• Continuous Learning System: Implementing a continuous learning mechanism will allow the system to improve over time by learning from user interactions and feedback, thereby enhancing accuracy and performance.

These improvements will help transform ResumeInsight into a more intelligent, scalable, and user-centric platform, making it highly valuable for job seekers across different domains.

CHAPTER 6: References

The following references were used during the development of the ResumeInsight project. These sources provided valuable information regarding Artificial Intelligence, Natural Language Processing, web development, and resume analysis techniques.

1. YouTube. Various tutorials on Python, Flask, AI integration, and web development concepts. Available at: https://www.youtube.com
2. CodeWithHarry. Educational resources and tutorials for Python programming, Flask framework, and web development. Available at: https://codewithharry.com
3. Kaggle. Datasets and learning resources related to machine learning, natural language processing, and data analysis. Available at: https://www.kaggle.com
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5. PyPDF2 Documentation. Python library used for extracting text from PDF files. Available at: https://pypi.org/project/PyPDF2/
6. python-docx Documentation. Library used for processing DOCX files. Available at: https://pypi.org/project/python-docx/
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8. OpenPyXL Documentation. Library used for creating and editing Excel files. Available at: https://openpyxl.readthedocs.io
9. Flask Documentation. Official documentation for Flask web framework. Available at: https://flask.palletsprojects.com
10. Google Gemini API Documentation. Used for AI-based resume analysis and content generation.
11. Chart.js Documentation. Used for data visualization and analytics dashboard. Available at: https://www.chartjs.org
12. Tailwind CSS Documentation. Used for designing responsive and modern user interfaces. Available at: https://tailwindcss.com/docs
13. SQLAlchemy Documentation. Used for database management and ORM. Available at: https://docs.sqlalchemy.org

CHAPTER 6:

Code: app.py

```
from flask import Flask,
render_template, request,
jsonify, send_file, session,
redirect, url_for
from flask_cors import
CORS
from flask_sqlalchemy
import SQLAlchemy
from werkzeug.utils import
secure_filename
import os
import json
import re
import sqlite3
from datetime import
datetime, timedelta
import io
from reportlab.lib.pagesizes
import letter, A4
from reportlab.platypus
import
SimpleDocTemplate, Table,
TableStyle, Paragraph,
Spacer, PageBreak, Image
from reportlab.lib.styles
import
```

```
getSampleStyleSheet,
ParagraphStyle
from reportlab.lib.units
import inch, cm
from reportlab.lib import
colors
from reportlab.pdfgen
import canvas
from reportlab.lib.enums
import TA_CENTER,
TA_LEFT, TA_RIGHT,
TA_JUSTIFY
import google.generativeai
as genai
import PyPDF2
import docx
from pathlib import Path
from functools import
wraps
import threading
import openpyxl
from openpyxl.styles
import Font, PatternFill,
Alignment
import base64
from io import BytesIO
from collections import
Counter
import statistics
```

```
from dotenv import
load_dotenv

# Load environment
variables
load_dotenv()

app = Flask(__name__)
CORS(app)

# Configuration
app.config['SECRET_KEY'
] = 'your-secret-key-
change-in-production'
app.config['SQLALCHEM
Y_DATABASE_URI'] =
'sqlite:///resume_analyzer.d
b'
app.config['MAX_CONTE
NT_LENGTH'] = 50 *
1024 * 1024
app.config['UPLOAD_FOL
DER'] = 'temp_uploads'
os.makedirs(app.config['UP
LOAD_FOLDER'],
exist_ok=True)

# Initialize extensions
db = SQLAlchemy(app)
```

```
# Initialize Gemini
def init_gemini():
    api_key =
os.getenv('GEMINI_API_
KEY')
    if not api_key:
        raise
ValueError("GEMINI_API
_KEY not found in
environment variables.
Please add it to your .env
file.")

genai.configure(api_key=a
pi_key)

init_gemini()

# Database Models
class Resume(db.Model):
    id =
db.Column(db.Integer,
primary_key=True)
    user_id =
db.Column(db.Integer,
default=1)
```

```
    filename =
db.Column(db.String(200),
nullable=False)
    text_content =
db.Column(db.Text)
    version =
db.Column(db.Integer,
default=1)
    version_name =
db.Column(db.String(100))
    parent_resume_id =
db.Column(db.Integer,
db.ForeignKey('resume.id')
)
    created_at =
db.Column(db.DateTime,
default=datetime.utcnow)
    analyses =
db.relationship('Analysis',
backref='resume',
lazy=True)

class Analysis(db.Model):
    id =
db.Column(db.Integer,
primary_key=True)
    user_id =
db.Column(db.Integer,
default=1)
```

```
    resume_id =
db.Column(db.Integer,
db.ForeignKey('resume.id')
)
    job_title =
db.Column(db.String(200))
    company =
db.Column(db.String(200))
    jd_text =
db.Column(db.Text)
    analysis_data =
db.Column(db.JSON)
    overall_score =
db.Column(db.Float)
    created_at =
db.Column(db.DateTime,
default=datetime.utcnow)

class
SkillTracker(db.Model):
    id =
db.Column(db.Integer,
primary_key=True)
    user_id =
db.Column(db.Integer,
default=1)
    skill_name =
db.Column(db.String(100))
```